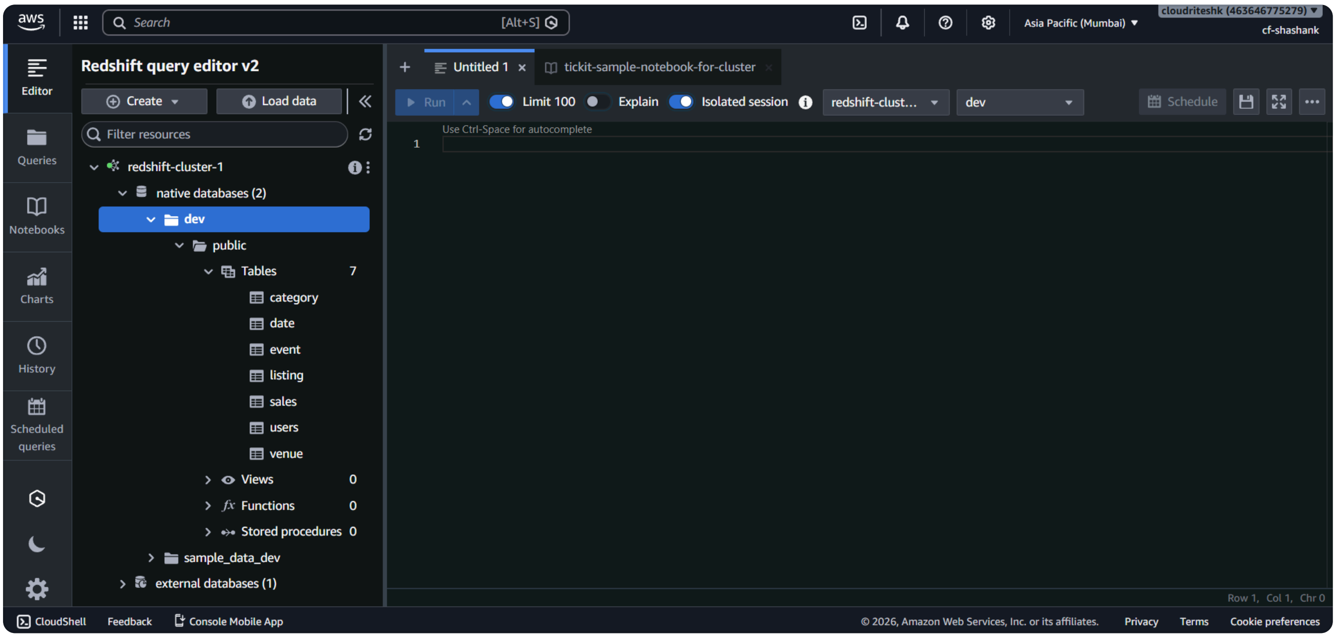
Demo: User Defined Function

Quick Summary

Creat a SQL-based UDF named `f_concat_ws` in Amazon Redshift to concatenate two strings with a space between them, verified the function inside the Functions section, and used it on the `users` table to combine first and last names successfully.

Environment Used

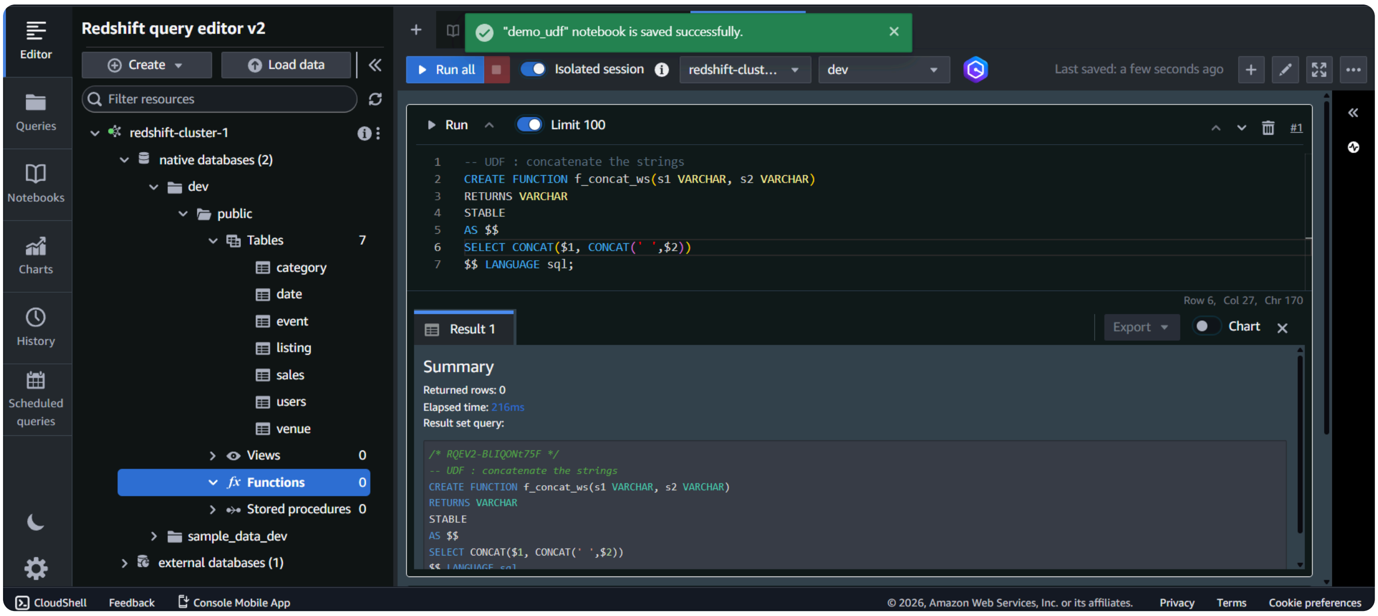
Worked inside:

- Redshift Query Editor v2
- Connected Redshift Cluster with Tickit sample database loaded
- 

Available sample tables:

- sales
- listing
- category
- users

Functions Section



On the left panel:

- Opened:
    - `Functions`

Initially: `0 Functions`

This means no custom UDFs were created yet.

Creating the UDF

Function Name

f_concat_ws

Purpose:

- Concatenate two strings with a space between them

UDF Logic

The function:

1. Takes two `VARCHAR` strings
2. Adds a space before the second string
3. Concatenates both strings together

Why nested CONCAT?

Because: `CONCAT()`

accepts only two parameters at a time.

So nested `CONCAT` operations were used.

SQL UDF Used

```
CREATE FUNCTION f_concat_ws(varchar, varchar)
```

Language used: SQL

Reason:

- Function logic was written completely in SQL.

Result After Creation

After running the command:

- Function created successfully.

Refreshing the left panel showed: `Functions: 1`

The function: `f_concat_ws` became visible inside the Functions section.

Using the UDF

Table Used

users

The `users` table contains:

- `firstname`
- `lastname`
- `username`

Query Performed

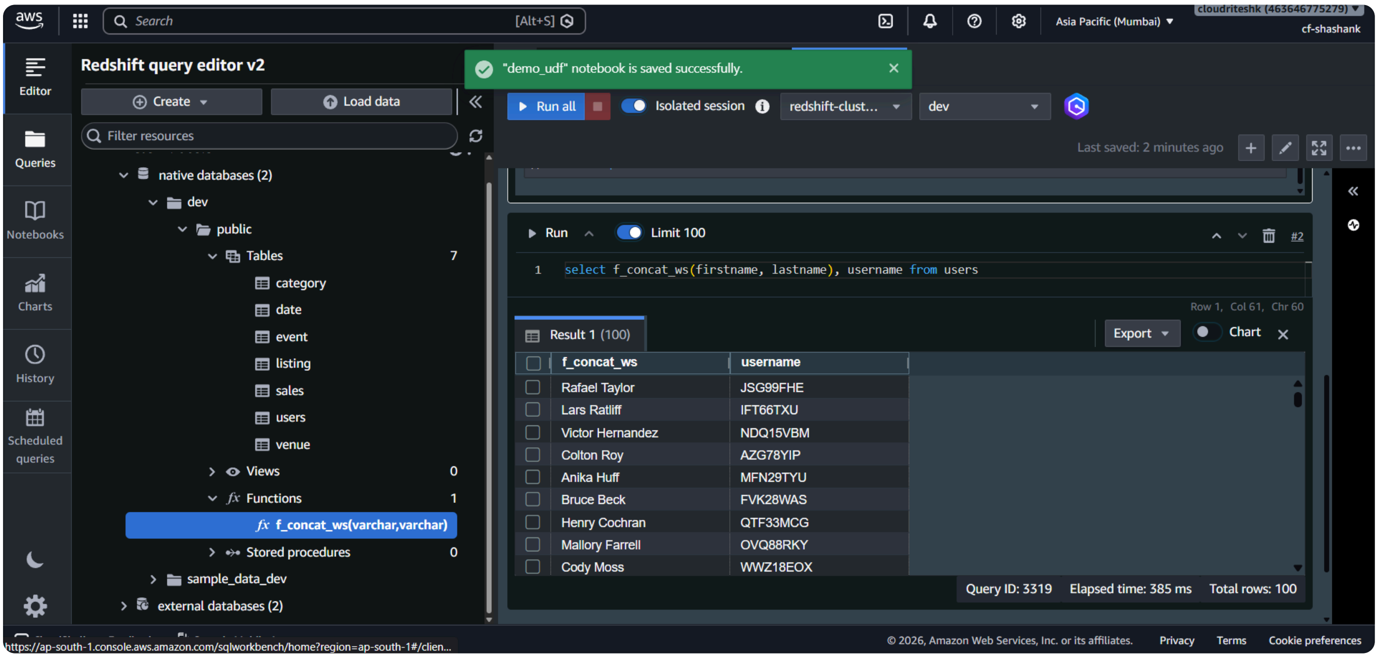
Used:

`f_concat_ws(firstname, lastname)`

Purpose:

- Merge first name and last name into a single value.

Output Received



The query returned:

- Username

- Full name with space between first and last name

Example output format: `John Smith`

instead of: `JohnSmith`

Key Learning

A UDF in Redshift helps:

- Reuse custom logic
- Simplify SQL queries
- Perform repeated transformations easily

UDFs can also be used for:

- Complex calculations
- String formatting
- Custom business logic
